

Overall MILP Problem Formulation and Methodology for Coordinated EV Fast-Charging Station Energy Management and Dynamic Tariff Generation

Methodology corresponding to the implemented Pyomo–HiGHS controller

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Abstract

This document presents the overall problem formulation and complete methodology of a mixed-integer linear programming controller for a grid-connected electric-vehicle fast-charging station integrated with photovoltaic generation and a battery energy storage system. The controller coordinates exact-minute EV-session scheduling, 15-minute power allocation, grid import and PV export, battery charging and discharging, and secondary-user tariff selection. Primary users retain fixed booked charging sessions, while all secondary users are served through whole-session shifting within user-type-specific time windows. A three-stage lexicographic procedure first maximizes direct PV supply to EV charging, then maximizes PV energy stored in the BESS while maintaining the first-stage result, and finally maximizes station profit while maintaining the renewable-energy priorities. The dynamic tariff is selected simultaneously with EV scheduling and energy dispatch from a discrete set of prices. Solar availability, price responsiveness, grid prices, export revenue, battery operation, curtailment, and shifting discomfort jointly influence the final tariff and dispatch schedule.

Contents

1	Problem Statement	2
2	System Configuration and Control Boundary	2
3	Overall Controller Workflow	3
4	Sets, Indices, and Time Resolution	4
5	Main Input Parameters	5

6	EV Session Preprocessing	5
6.1	Required energy	5
6.2	Session duration and power	6
6.3	Whole-session candidate starts	6
6.4	Fractional overlap with 15-minute slots	6
7	Decision Variables	6
7.1	EV start selection	6
7.2	Tariff selection	6
7.3	Tariff-linked secondary demand	7
7.4	Energy-flow and state variables	7
8	EV Load Formulation	7
9	EV Scheduling and Physical Charger Constraints	8
9.1	Every EV is served exactly once	8
9.2	Maximum number of shifted secondary users	8
9.3	One EV per charger at an exact minute	8
9.4	Exact-minute station constraints	8
9.5	Slot-level station and load-ramp constraints	9
10	Dynamic Secondary Tariff Formulation	9
10.1	One tariff in every slot	9
10.2	Link between tariff and secondary load	9
10.3	Potential secondary demand	9
10.4	Price-attraction relationship	9
10.5	Tariff-ramp constraint	10
10.6	Solar score and reference tariff	10
11	Energy-Management Constraints	10
11.1	EV power balance	10
11.2	PV power balance	10
12	Grid Import and Export Constraints	11
13	BESS Operating Constraints	11
14	Three-Stage Lexicographic Objective	12
14.1	Stage 1: maximize direct PV supply to EVs	12
14.2	Stage 2: maximize PV charging of the BESS	12
14.3	Stage 3: maximize station profit and operating value	13
14.3.1	Primary-user revenue	13
14.3.2	Secondary-user revenue	13
14.3.3	PV export revenue	13
14.3.4	Grid import cost	13
14.3.5	Battery degradation cost	13

14.3.6 PV-curtailment cost	13
14.3.7 Solar-price alignment cost	14
14.3.8 Solar-shifting reward	14
14.3.9 Non-solar secondary penalty	14
14.3.10 Shifting discomfort cost	14
15 How Tariff, EV Shifting, and Energy Dispatch Are Coordinated	14
16 Solution Algorithm	15
17 Main Model Outputs	16
17.1 Slot-level outputs	16
17.2 User-level outputs	16
17.3 Physical validation outputs	16
18 Methodological Interpretation	17
19 Important Modeling Assumptions	17
20 Conclusion	18

1 Problem Statement

A fast-charging station can experience high and rapidly varying electrical demand because several high-power EV chargers may operate simultaneously. When the station is connected to a PV plant and a BESS, the controller must decide how the EV demand is supplied in every operating interval. It must also decide when flexible charging sessions should occur and what selling tariff should be offered to secondary users.

The optimization problem is therefore to determine a feasible daily operating schedule that:

1. serves every primary and secondary EV charging session;
2. preserves fixed charging times for booked primary users;
3. shifts only eligible secondary-user sessions and keeps each session continuous;
4. maximizes the direct use of PV energy by EV charging;
5. stores available PV energy in the BESS before unnecessary export or curtailment;
6. respects charger, station, grid, BESS, and ramp-rate limits;
7. prevents simultaneous import and export and simultaneous BESS charging and discharging;
8. selects one secondary-user tariff in every 15-minute interval; and
9. maximizes the daily economic benefit after preserving the PV-utilization priorities.

The formulation is a MILP because it contains continuous power and energy variables together with binary variables for EV start selection, tariff selection, grid operating mode, and BESS operating mode.

2 System Configuration and Control Boundary

The modeled station consists of:

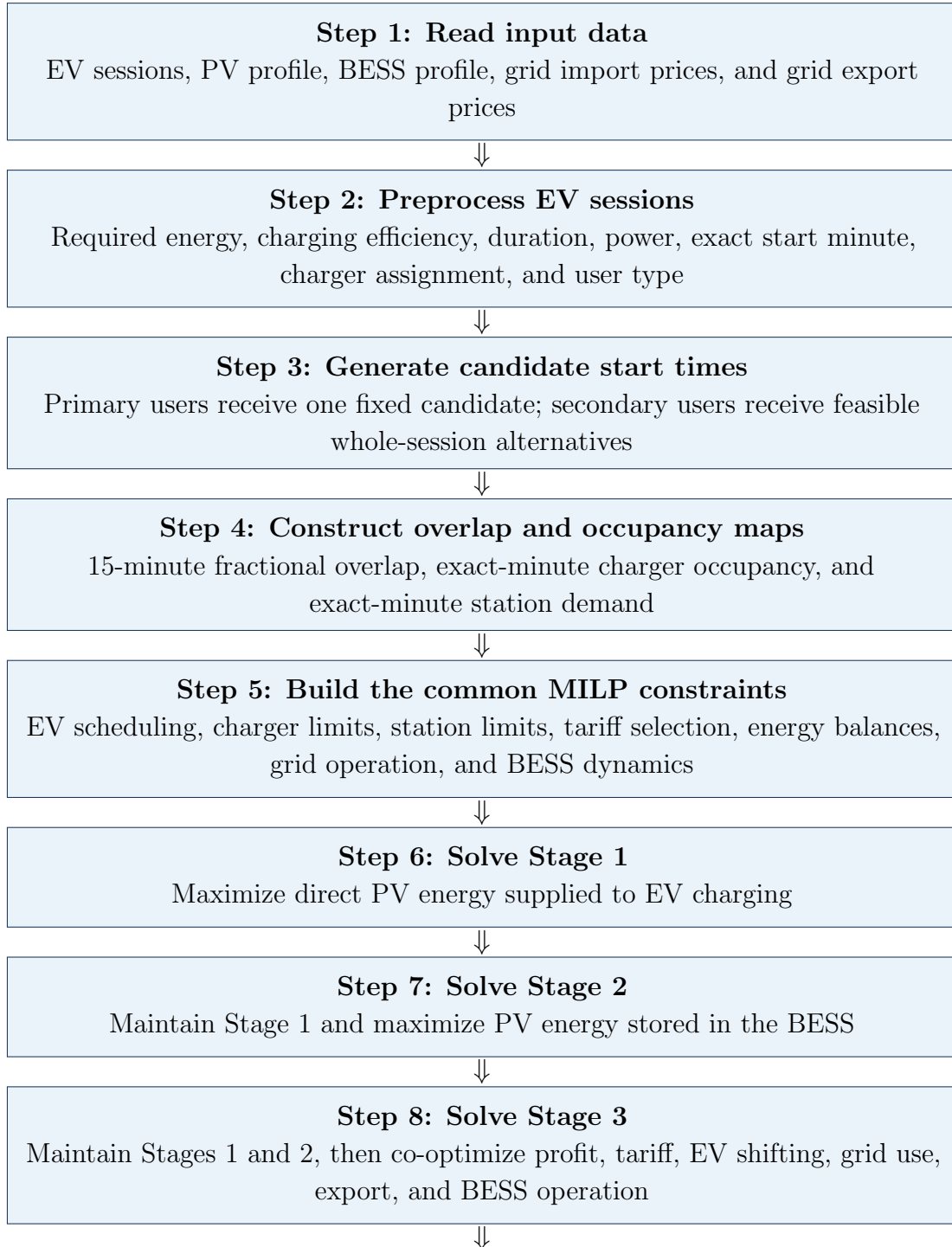
- ten DC fast-charging piles, each rated at 450 kW;
- a 5 MW PV generation system;
- a 5200 kWh BESS with a 2600 kW charge/discharge rating;
- a grid connection with a 4000 kW import limit;
- primary EV users with fixed booked charging times;
- secondary EV users classified as opportunistic, elastic, or long-trip users;
- time-varying 15-minute grid import and export prices; and

- a discrete set of secondary-user selling tariffs from 50 to 150 LKR/kWh.

The optimization horizon is one day. Energy management is evaluated in 96 intervals of 15 minutes, while charger occupancy and instantaneous station power are checked at one-minute resolution.

3 Overall Controller Workflow

The complete controller follows the sequence shown below.



Step 9: Extract and validate results

EV start times, tariffs, power flows, SOC, profit, charger occupancy, ramp compliance, and output files

The three MILP stages use the same physical constraints and decision-variable structure. Only the objective function and the minimum retained values from previous stages change.

4 Sets, Indices, and Time Resolution

Symbol	Definition
$\mathcal{T}$	Set of 15-minute time slots, $\mathcal{T} = \{0, 1, \dots, 95\}$.
$\mathcal{M}$	Set of exact minutes, $\mathcal{M} = \{0, 1, \dots, 1439\}$.
$\mathcal{U}$	Set of all EV charging sessions.
$\mathcal{U}^{\text{pri}}$	Set of primary EV users.
$\mathcal{U}^{\text{sec}}$	Set of secondary EV users.
$\mathcal{S}_u$	Set of feasible candidate start slots for EV u .
$\mathcal{K}$	Set of available secondary tariff levels.
$\mathcal{C}$	Set of charger piles, $\mathcal{C} = \{1, \dots, 10\}$.
t	15-minute slot index.
m	Exact-minute index.
u	EV-session index.
s	Candidate start-slot index.
k	Tariff-level index.
c	Charger-pile index.

The interval duration is

$$\Delta t = 0.25 \text{ h.} \quad (1)$$

5 Main Input Parameters

Table 2: Principal controller parameters.

Parameter	Symbol	Value
Number of time slots	N	96
Number of charger piles	N_c	10
Charger-pile power limit	$P_c^{\max}$	450 kW
Station charging capacity	$P_{st}^{\max}$	4500 kW
Grid import limit	$P_{Grid}^{\max}$	4000 kW
PV rated power	P_{PV}^{rated}	5000 kW
BESS energy capacity	E_{BESS}^{rated}	5200 kWh
BESS charge/discharge limit	$P_{BESS}^{\max}$	2600 kW
BESS charging efficiency	η_{ch}	0.95
BESS discharging efficiency	η_{dis}	0.95
Minimum SOC	$SOC^{\min}$	520 kWh
Maximum SOC	$SOC^{\max}$	4680 kWh
Initial SOC	SOC^0	2600 kWh
EV-load ramp limit	$R_{EV}^{\max}$	1000 kW/slot
Grid-import ramp limit	$R_{Grid}^{\max}$	1000 kW/slot
Tariff ramp limit	$R_C^{\max}$	20
		LKR/kWh/slot
PV export limit	$P_{exp}^{\max}$	4000 kW
Maximum shifted-secondary fraction	$\rho_{shift}^{\max}$	45%
PV-curtailment penalty	c_{curt}	50 LKR/kWh
Solar-shift reward coefficient	c_{solar}	150 LKR/kWh
Non-solar secondary penalty	c_{non}	35 LKR/kWh
Shift discomfort cost	c_{shift}	140 LKR/session
Solar-price alignment coefficient	c_{align}	120 LKR/kWh

The time-varying grid import price and grid export price are denoted by $c_t^{\text{Grid,buy}}$ and $c_t^{\text{Grid,sell}}$, respectively.

6 EV Session Preprocessing

6.1 Required energy

For EV u , the battery-side energy requirement can be calculated from the battery capacity and the requested SOC change:

$$E_u^{\text{req}} = E_u^{\text{bat}} \frac{SOC_u^{\text{target}} - SOC_u^{\text{entry}}}{100}. \quad (2)$$

If the input file provides a requested or measured energy value, that value is used directly.

With charging efficiency η_u^{EV} , the required charger-side energy is

$$E_u^{\text{ch}} = \frac{E_u^{\text{req}}}{\eta_u^{\text{EV}}}. \quad (3)$$

6.2 Session duration and power

Let D_u be the charging-session duration in minutes. The average session power is

$$P_u = \min \left(\frac{60E_u^{\text{ch}}}{D_u}, P_u^{\text{EV,max}}, P_u^{\text{controller}}, P_c^{\text{max}} \right). \quad (4)$$

The duration is increased when necessary to ensure that the required energy can be delivered without exceeding the power limit.

6.3 Whole-session candidate starts

Each EV receives a candidate start set $\mathcal{S}_u$.

For a primary user,

$$\mathcal{S}_u = \{s_u^0\}, \quad u \in \mathcal{U}^{\text{pri}}, \quad (5)$$

where s_u^0 is the booked start slot.

For a secondary user, $\mathcal{S}_u$ contains the original start and feasible shifted starts within the user-type-specific scheduling window. Opportunistic and elastic users may be shifted inside the main secondary window, while long-trip users are allowed a longer latest-start range.

6.4 Fractional overlap with 15-minute slots

Because a session can begin at any exact minute, its contribution to a 15-minute slot may be fractional. Let $b_{u,s}$ and $e_{u,s}$ be the candidate start and end minutes. The overlap fraction is

$$\alpha_{u,s,t} = \frac{\max(0, \min(e_{u,s}, 15(t+1)) - \max(b_{u,s}, 15t))}{15}. \quad (6)$$

Hence,

$$0 \leq \alpha_{u,s,t} \leq 1. \quad (7)$$

7 Decision Variables

7.1 EV start selection

$$X_{u,s} = \begin{cases} 1, & \text{if EV } u \text{ starts at candidate slot } s, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

7.2 Tariff selection

$$z_{t,k} = \begin{cases} 1, & \text{if tariff level } C_k \text{ is selected in slot } t, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

The tariff levels are

$$\mathcal{K} = \{50, 55, 60, \dots, 145, 150\} \text{ LKR/kWh.} \quad (10)$$

The selected tariff is

$$C_t^{\text{sec}} = \sum_{k \in \mathcal{K}} C_k z_{t,k}. \quad (11)$$

7.3 Tariff-linked secondary demand

$P_{t,k}^{\text{sec}}$ is the secondary EV load associated with tariff level k in slot t . This auxiliary variable is used to calculate tariff revenue without multiplying two decision variables.

7.4 Energy-flow and state variables

The principal continuous variables are:

$$P_t^{\text{PV,EV}} : \text{PV power supplied directly to EVs,} \quad (12)$$

$$P_t^{\text{PV,BESS}} : \text{PV power used to charge the BESS,} \quad (13)$$

$$P_t^{\text{PV,exp}} : \text{PV power exported to the grid,} \quad (14)$$

$$P_t^{\text{PV,curt}} : \text{curtailed PV power,} \quad (15)$$

$$P_t^{\text{Grid,EV}} : \text{grid power supplied to EVs,} \quad (16)$$

$$P_t^{\text{Grid,BESS}} : \text{grid power used to charge the BESS,} \quad (17)$$

$$P_t^{\text{BESS,EV}} : \text{BESS discharge power supplied to EVs,} \quad (18)$$

$$SOC_t : \text{BESS stored energy at the end of slot } t. \quad (19)$$

The binary grid-mode variable is y_t^{Grid} , and the binary BESS-mode variable is y_t^{BESS} .

8 EV Load Formulation

The primary-user charging load in slot t is

$$P_t^{\text{pri}} = \sum_{u \in \mathcal{U}^{\text{pri}}} \sum_{s \in \mathcal{S}_u} P_u \alpha_{u,s,t} X_{u,s}. \quad (20)$$

The secondary-user charging load is

$$P_t^{\text{sec}} = \sum_{u \in \mathcal{U}^{\text{sec}}} \sum_{s \in \mathcal{S}_u} P_u \alpha_{u,s,t} X_{u,s}. \quad (21)$$

The total EV demand is

$$P_t^{\text{EV}} = P_t^{\text{pri}} + P_t^{\text{sec}}. \quad (22)$$

9 EV Scheduling and Physical Charger Constraints

9.1 Every EV is served exactly once

Each EV must select one complete session start:

$$\sum_{s \in \mathcal{S}_u} X_{u,s} = 1, \quad \forall u \in \mathcal{U}. \quad (23)$$

Because a primary user has only one candidate start, Equation (23) fixes that user at the booked time. Every secondary EV is also served, but its complete session can be shifted among its feasible candidates.

9.2 Maximum number of shifted secondary users

Let s_u^0 be the original start of secondary EV u . The total number of shifted secondary users is constrained by

$$\sum_{u \in \mathcal{U}^{\text{sec}}} \sum_{\substack{s \in \mathcal{S}_u \\ s \neq s_u^0}} X_{u,s} \leq N_{\text{shift}}^{\text{max}}, \quad (24)$$

where

$$N_{\text{shift}}^{\text{max}} = \lfloor \rho_{\text{shift}}^{\text{max}} |\mathcal{U}^{\text{sec}}| \rfloor. \quad (25)$$

9.3 One EV per charger at an exact minute

Let $\mathcal{A}_{c,m}$ be the candidate sessions that would use charger c at minute m . Then

$$\sum_{(u,s) \in \mathcal{A}_{c,m}} X_{u,s} \leq 1, \quad \forall c, m. \quad (26)$$

The exact-minute power on a charger is constrained by

$$\sum_{(u,s) \in \mathcal{A}_{c,m}} P_u X_{u,s} \leq P_c^{\text{max}}, \quad \forall c, m. \quad (27)$$

9.4 Exact-minute station constraints

Let $\mathcal{A}_m$ contain all candidate sessions active at minute m . The number of simultaneous EVs is

$$\sum_{(u,s) \in \mathcal{A}_m} X_{u,s} \leq N_c, \quad \forall m. \quad (28)$$

The total instantaneous station charging power satisfies

$$\sum_{(u,s) \in \mathcal{A}_m} P_u X_{u,s} \leq P_{\text{st}}^{\text{max}}, \quad \forall m. \quad (29)$$

9.5 Slot-level station and load-ramp constraints

$$P_t^{\text{EV}} \leq P_{\text{st}}^{\text{max}}, \quad \forall t. \quad (30)$$

The EV demand change between consecutive slots is limited by

$$P_t^{\text{EV}} - P_{t-1}^{\text{EV}} \leq R_{\text{EV}}^{\text{max}}, \quad (31)$$

$$P_{t-1}^{\text{EV}} - P_t^{\text{EV}} \leq R_{\text{EV}}^{\text{max}}, \quad \forall t > 0. \quad (32)$$

10 Dynamic Secondary Tariff Formulation

10.1 One tariff in every slot

$$\sum_{k \in \mathcal{K}} z_{t,k} = 1, \quad \forall t. \quad (33)$$

10.2 Link between tariff and secondary load

The total secondary load is allocated to the tariff-linked variables:

$$P_t^{\text{sec}} = \sum_{k \in \mathcal{K}} P_{t,k}^{\text{sec}}, \quad \forall t. \quad (34)$$

Only a selected tariff may carry secondary demand:

$$P_{t,k}^{\text{sec}} \leq P_{\text{st}}^{\text{max}} z_{t,k}, \quad \forall t, k. \quad (35)$$

If $z_{t,k} = 0$, Equation (35) forces $P_{t,k}^{\text{sec}} = 0$.

10.3 Potential secondary demand

Let $\mathcal{A}_{u,t}^{\text{sec}}$ be the candidate starts of secondary EV u that overlap slot t . The maximum possible contribution of EV u to that slot is

$$\bar{P}_{u,t}^{\text{sec}} = \max_{s \in \mathcal{A}_{u,t}^{\text{sec}}} (P_u \alpha_{u,s,t}). \quad (36)$$

The total physically available secondary power is

$$\bar{P}_t^{\text{sec}} = \sum_{u \in \mathcal{U}^{\text{sec}}} \bar{P}_{u,t}^{\text{sec}}. \quad (37)$$

Each physical EV is represented once in Equation (37), even when several of its candidate start options overlap the same slot.

10.4 Price-attraction relationship

Each tariff level C_k has a fixed attraction factor a_k . Lower prices have larger factors and higher prices have smaller factors. The secondary load at the selected tariff is constrained

by

$$P_{t,k}^{\text{sec}} \leq \bar{P}_t^{\text{sec}} a_k z_{t,k}, \quad \forall t, k. \quad (38)$$

The implemented attraction factors decrease from 1.50 at 50 LKR/kWh to 0.12 at 150 LKR/kWh. Therefore, a high tariff can be selected only when the required slot demand is compatible with its lower attraction factor.

10.5 Tariff-ramp constraint

The selected tariff cannot change by more than 20 LKR/kWh between adjacent slots:

$$C_t^{\text{sec}} - C_{t-1}^{\text{sec}} \leq R_C^{\text{max}}, \quad (39)$$

$$C_{t-1}^{\text{sec}} - C_t^{\text{sec}} \leq R_C^{\text{max}}, \quad \forall t > 0. \quad (40)$$

10.6 Solar score and reference tariff

The available PV surplus relative to the primary load is converted into a normalized solar score:

$$S_t = \min \left[\frac{\max(P_t^{\text{PV}} - P_t^{\text{pri},0}, 0)}{P_{\text{solar}}^{\text{ref}}}, 1 \right], \quad (41)$$

where $P_t^{\text{pri},0}$ is the reference primary load and $P_{\text{solar}}^{\text{ref}} = 1000$ kW.

A solar-based reference tariff is defined as

$$C_t^{\text{target}} = C^{\text{max}} - S_t(C^{\text{max}} - C^{\text{min}}), \quad (42)$$

with $C^{\text{min}} = 50$ and $C^{\text{max}} = 150$ LKR/kWh.

Thus, high solar availability produces a lower reference tariff, while low solar availability produces a higher reference tariff. This is a soft economic reference used in the Stage 3 objective; it does not prohibit any tariff level.

11 Energy-Management Constraints

11.1 EV power balance

In every slot, the EV charging demand must be supplied by PV, the grid, and BESS discharge:

$$P_t^{\text{PV,EV}} + P_t^{\text{Grid,EV}} + P_t^{\text{BESS,EV}} = P_t^{\text{EV}}, \quad \forall t. \quad (43)$$

11.2 PV power balance

All PV generation is allocated among direct EV supply, BESS charging, export, and curtailment:

$$P_t^{\text{PV,EV}} + P_t^{\text{PV,BESS}} + P_t^{\text{PV,exp}} + P_t^{\text{PV,curt}} = P_t^{\text{PV}}, \quad \forall t. \quad (44)$$

12 Grid Import and Export Constraints

The total grid import is

$$P_t^{\text{Grid}} = P_t^{\text{Grid,EV}} + P_t^{\text{Grid,BESS}}. \quad (45)$$

The grid-import limit is

$$P_t^{\text{Grid}} \leq P_{\text{Grid}}^{\text{max}}. \quad (46)$$

A binary variable y_t^{Grid} prevents simultaneous grid import and PV export:

$$P_t^{\text{Grid}} \leq P_{\text{Grid}}^{\text{max}} y_t^{\text{Grid}}, \quad (47)$$

$$P_t^{\text{PV,exp}} \leq P_{\text{exp}}^{\text{max}} (1 - y_t^{\text{Grid}}). \quad (48)$$

Therefore, $y_t^{\text{Grid}} = 1$ permits import and blocks export, while $y_t^{\text{Grid}} = 0$ permits export and blocks import.

The grid-import ramp is restricted by

$$P_t^{\text{Grid}} - P_{t-1}^{\text{Grid}} \leq R_{\text{Grid}}^{\text{max}}, \quad (49)$$

$$P_{t-1}^{\text{Grid}} - P_t^{\text{Grid}} \leq R_{\text{Grid}}^{\text{max}}, \quad \forall t > 0. \quad (50)$$

Grid-to-BESS charging is permitted only during the specified off-peak period, 22:30–05:30. Outside that period,

$$P_t^{\text{Grid,BESS}} = 0. \quad (51)$$

13 BESS Operating Constraints

The BESS charging power is

$$P_t^{\text{ch}} = P_t^{\text{PV,BESS}} + P_t^{\text{Grid,BESS}}, \quad (52)$$

and its discharging power is

$$P_t^{\text{dis}} = P_t^{\text{BESS,EV}}. \quad (53)$$

The power limits are

$$P_t^{\text{ch}} \leq P_{\text{BESS}}^{\text{max}}, \quad (54)$$

$$P_t^{\text{dis}} \leq P_{\text{BESS}}^{\text{max}}. \quad (55)$$

The binary BESS mode prevents simultaneous charging and discharging:

$$P_t^{\text{ch}} \leq P_{\text{BESS}}^{\text{max}} y_t^{\text{BESS}}, \quad (56)$$

$$P_t^{\text{dis}} \leq P_{\text{BESS}}^{\text{max}} (1 - y_t^{\text{BESS}}). \quad (57)$$

For the first slot, the SOC equation is

$$SOC_0 = SOC^0 + \eta_{\text{ch}} P_0^{\text{ch}} \Delta t - \frac{P_0^{\text{dis}} \Delta t}{\eta_{\text{dis}}}. \quad (58)$$

For $t > 0$,

$$SOC_t = SOC_{t-1} + \eta_{\text{ch}} P_t^{\text{ch}} \Delta t - \frac{P_t^{\text{dis}} \Delta t}{\eta_{\text{dis}}}. \quad (59)$$

The SOC bounds are

$$SOC^{\min} \leq SOC_t \leq SOC^{\max}, \quad \forall t. \quad (60)$$

The terminal condition is

$$SOC_{95} \geq SOC^0. \quad (61)$$

This prevents the daily optimization from increasing profit simply by ending the day with a depleted battery.

14 Three-Stage Lexicographic Objective

The controller solves three MILPs sequentially. The result of each earlier stage is retained as a lower-bound constraint in the next stage.

14.1 Stage 1: maximize direct PV supply to EVs

$$\boxed{\max J_1 = \sum_{t \in \mathcal{T}} P_t^{\text{PV}, \text{EV}} \Delta t} \quad (62)$$

Let the optimal Stage 1 value be

$$E_{\text{PV} \rightarrow \text{EV}}^{\max}. \quad (63)$$

This stage gives the highest priority to direct use of PV energy by EV charging.

14.2 Stage 2: maximize PV charging of the BESS

Stage 2 preserves the Stage 1 result:

$$\sum_t P_t^{\text{PV}, \text{EV}} \Delta t \geq E_{\text{PV} \rightarrow \text{EV}}^{\max} - \varepsilon, \quad (64)$$

where $\varepsilon = 1$ kWh.

The Stage 2 objective is

$$\boxed{\max J_2 = 1000 \sum_t P_t^{\text{PV}, \text{BESS}} \Delta t + \sum_t c_t^{\text{Grid}, \text{sell}} P_t^{\text{PV}, \text{exp}} \Delta t - 200 \sum_t P_t^{\text{PV}, \text{curt}} \Delta t} \quad (65)$$

The large coefficient 1000 gives first priority within Stage 2 to PV-to-BESS energy. Export revenue is then rewarded, and PV curtailment is penalized.

Let the resulting PV-to-BESS energy be

$$E_{\text{PV} \rightarrow \text{BESS}}^{\max}. \quad (66)$$

14.3 Stage 3: maximize station profit and operating value

Stage 3 preserves both renewable-energy results:

$$\sum_t P_t^{\text{PV},\text{EV}} \Delta t \geq E_{\text{PV} \rightarrow \text{EV}}^{\max} - \varepsilon, \quad (67)$$

$$\sum_t P_t^{\text{PV},\text{BESS}} \Delta t \geq E_{\text{PV} \rightarrow \text{BESS}}^{\max} - \varepsilon. \quad (68)$$

The final objective is

$$\boxed{\max J_3 = R_{\text{pri}} + R_{\text{sec}} + R_{\text{exp}} - C_{\text{Grid}} - C_{\text{deg}} - C_{\text{curt}} - C_{\text{align}} + R_{\text{solar}} - C_{\text{non}} - C_{\text{shift}}} \quad (69)$$

The individual terms are defined below.

14.3.1 Primary-user revenue

$$R_{\text{pri}} = \sum_t c_t^{\text{pri}} P_t^{\text{pri}} \Delta t, \quad (70)$$

where c_t^{pri} is the fixed primary-user time-of-use tariff.

14.3.2 Secondary-user revenue

$$R_{\text{sec}} = \sum_t \sum_k C_k P_{t,k}^{\text{sec}} \Delta t. \quad (71)$$

Because C_k is a fixed parameter, Equation (71) is linear.

14.3.3 PV export revenue

$$R_{\text{exp}} = \sum_t c_t^{\text{Grid,sell}} P_t^{\text{PV,exp}} \Delta t. \quad (72)$$

14.3.4 Grid import cost

$$C_{\text{Grid}} = \sum_t c_t^{\text{Grid,buy}} \left(P_t^{\text{Grid,EV}} + P_t^{\text{Grid,BESS}} \right) \Delta t. \quad (73)$$

14.3.5 Battery degradation cost

$$C_{\text{deg}} = \sum_t c_{\text{deg}} P_t^{\text{BESS,EV}} \Delta t. \quad (74)$$

The implemented coefficient is currently zero, so the term is available in the formulation but does not affect the present solution.

14.3.6 PV-curtailment cost

$$C_{\text{curt}} = \sum_t c_{\text{curt}} P_t^{\text{PV,curt}} \Delta t. \quad (75)$$

14.3.7 Solar-price alignment cost

The deviation of each tariff level from the slot reference tariff is normalized by the total tariff range. The alignment cost is

$$C_{\text{align}} = \sum_t \sum_k c_{\text{align}} \frac{|C_k - C_t^{\text{target}}|}{C_{\text{max}} - C_{\text{min}}} P_{t,k}^{\text{sec}} \Delta t. \quad (76)$$

The absolute value in Equation (76) is applied only to fixed parameter values, so every coefficient is known before optimization and the model remains linear. The cost applies only to secondary energy actually delivered. It guides the tariff toward lower values during solar-rich periods and higher values during low-solar periods without imposing a hard tariff cap.

14.3.8 Solar-shifting reward

Let w_u be the user-type solar-shift weight: 1.0 for opportunistic, 1.8 for elastic, and 0.6 for long-trip users. The solar reward is

$$R_{\text{solar}} = \sum_t \sum_{u \in \mathcal{U}^{\text{sec}}} \sum_{s \in S_u} c_{\text{solar}} w_u S_t P_u \alpha_{u,s,t} X_{u,s} \Delta t. \quad (77)$$

14.3.9 Non-solar secondary penalty

$$C_{\text{non}} = \sum_t \sum_{u \in \mathcal{U}^{\text{sec}}} \sum_{s \in S_u} c_{\text{non}} w_u (1 - S_t) P_u \alpha_{u,s,t} X_{u,s} \Delta t. \quad (78)$$

14.3.10 Shifting discomfort cost

$$C_{\text{shift}} = \sum_{u \in \mathcal{U}^{\text{sec}}} \sum_{\substack{s \in S_u \\ s \neq s_u^0}} c_{\text{shift}} X_{u,s}. \quad (79)$$

This term discourages unnecessary movement of secondary charging sessions.

15 How Tariff, EV Shifting, and Energy Dispatch Are Coordinated

The Stage 3 solver does not first calculate the tariff and then perform the energy dispatch. It simultaneously selects:

- one complete start time for every EV;
- one secondary tariff for every 15-minute slot;
- PV-to-EV, PV-to-BESS, export, and curtailment power;
- grid-to-EV and grid-to-BESS power;

- BESS charging, discharging, and SOC; and
- binary import/export and charge/discharge modes.

For each candidate solution, the optimizer evaluates the resulting revenue, grid cost, export revenue, solar usage, price-attraction compatibility, solar-price alignment, shifting reward, and user-discomfort cost. The selected solution is the feasible combination with the maximum Stage 3 objective value while retaining the Stage 1 and Stage 2 PV-energy results.

The expected economic behavior is:

high solar availability $\Rightarrow$ greater incentive to schedule secondary charging and select a lower tariff, (80)

and

low solar or expensive grid supply $\Rightarrow$ greater incentive for a higher tariff or a different secondary sched (81)

All secondary EVs remain served because Equation (23) is imposed for every EV.

16 Solution Algorithm

The implementation can be summarized as Algorithm 3.

Table 3: Sequential MILP solution algorithm.

Step	Operation
1	Read EV-session, PV, BESS, and dynamic grid-price inputs.
2	Convert all daily profiles to 96 slots and validate dimensions and prices.
3	Compute EV required energy, charger-side energy, duration, power, and exact-minute timing.
4	Generate candidate whole-session starts for each user class.
5	Build 15-minute overlap maps and exact-minute charger/station occupancy maps.
6	Compute the reference primary load, solar score, potential secondary power, and solar reference tariff.
7	Build and solve Stage 1; store $E_{PV \rightarrow EV}^{\max}$.
8	Build Stage 2 with the Stage 1 retention constraint; solve and store $E_{PV \rightarrow BESS}^{\max}$.
9	Build Stage 3 with both retention constraints; co-optimize tariff, shifting, dispatch, and profit.
10	Extract the selected start of each EV, selected tariff of each slot, all power flows, and BESS SOC.
11	Validate exact-minute charger occupancy, charger power, station demand, SOC, and ramp limits.
12	Export slot, user, charger, station, price, and summary results to CSV, Excel, and graph files.

The HiGHS MILP solver is used through Pyomo. The model accepts an optimal solution, a feasible solution, or the best available solution at the time limit, subject to the configured relative MIP gap.

17 Main Model Outputs

The controller produces the following result groups.

17.1 Slot-level outputs

- primary, secondary, and total EV demand;
- selected primary and secondary tariffs;
- PV generation and its allocation to EVs, BESS, export, and curtailment;
- grid-to-EV, grid-to-BESS, and total grid import;
- BESS charge, discharge, net power, and SOC;
- grid import and export prices;
- load, tariff, and grid-import ramps; and
- slot profit.

17.2 User-level outputs

- original and optimized charging start and end times;
- assigned charger pile;
- shifted duration and shifted-status indicator;
- delivered charger-side and battery-side energy;
- final EV SOC and charging cost;
- PV, grid, and BESS energy shares; and
- quality-of-service confirmation.

17.3 Physical validation outputs

- one-user-per-charger compliance at each exact minute;
- charger-pile power-limit compliance;
- maximum exact-minute station EV count;

- exact-minute station-power compliance;
- BESS SOC-limit compliance;
- final SOC compliance; and
- EV-load, tariff, and grid-ramp compliance.

18 Methodological Interpretation

The formulation combines three control levels within one daily optimization framework:

1. **Service scheduling:** primary bookings are preserved, and complete secondary sessions are shifted without interrupting their charging continuity.
2. **Energy management:** PV, BESS, and grid energy are dispatched to satisfy every EV load while respecting physical limits.
3. **Economic control:** the secondary tariff is selected together with dispatch and scheduling, using revenue, dynamic grid cost, export income, solar conditions, customer-attraction factors, and shifting costs.

The lexicographic sequence prevents the economic objective from sacrificing the main renewable-energy objectives. Direct PV supply to EVs remains the first priority, PV storage remains the second priority, and profit optimization is performed only after these two operating targets have been secured.

19 Important Modeling Assumptions

The methodology is based on the following assumptions:

- PV generation and grid prices are known for the optimization day.
- Each EV session has known energy, power, duration, efficiency, user type, and assigned charger.
- Primary users cannot be shifted.
- Every secondary user must be served exactly once.
- A charging session is continuous and is shifted as a complete block.
- The tariff is selected from a discrete set rather than as a fully continuous variable.
- Price-attraction factors are externally specified model parameters.
- BESS charging and discharging efficiencies are constant.
- No simultaneous grid import/export or BESS charge/discharge is allowed.
- Battery degradation cost is available in the model but is currently set to zero.

20 Conclusion

The developed MILP controller provides a unified method for daily EV-session scheduling, renewable-energy utilization, BESS management, grid interaction, and dynamic tariff generation. Its principal methodological feature is the three-stage lexicographic structure. Stage 1 secures the maximum direct PV supply to EV charging. Stage 2 stores the maximum available PV energy in the BESS while maintaining Stage 1. Stage 3 then selects the most profitable feasible combination of tariffs, secondary-user schedules, grid import, PV export, BESS operation, and curtailment while maintaining both earlier renewable-energy results.

The resulting tariff is therefore not produced independently from the physical energy schedule. It is generated as part of the same Stage 3 MILP solution that determines EV shifting and station energy dispatch. This provides a coordinated controller in which service obligations, charger limits, renewable-energy priorities, grid constraints, customer-price response, and station economics are considered together.